

Experiment 35:

Consider a file system that brings all the file pointers together into an index block. The *i*th entry in the index block points to the *i*th block of the file. Design a C program to simulate the file allocation strategy.

Code: -

```
#include <stdio.h>

#include <stdlib.h>

struct Block

{

    int blockNumber;

    char data[256];

};

int main()

{

    FILE *file;

    struct Block block;

    int blockNumber;

    long indexBlock[100] = {0};

    file = fopen("indexed_file.txt", "w");

    if (file == NULL)

    {

        printf("Error opening the file.\n");

        return 1;

    }
```

```
printf("Enter blocks (Enter '0' as block number to exit):\n");
```

```
while (1)
```

```
{
```

```
    printf("Block Number: ");
```

```
    scanf("%d", &block.blockNumber);
```

```
    if (block.blockNumber == 0)
```

```
    {
```

```
        break;
```

```
    }
```

```
    printf("Data: ");
```

```
    scanf(" %[^\n]", block.data);
```

```
    fwrite(&block, sizeof(struct Block), 1, file);
```

```
    indexBlock[block.blockNumber] =
```

```
        ftell(file) - sizeof(struct Block);
```

```
}
```

```
fclose(file);
```

```
file = fopen("indexed_file.txt", "r");
```

```
if (file == NULL)
```

```
{
```

```
    printf("Error opening the file.\n");
```

```
    return 1;
```

```
}
```

```
while (1)
```

```
{  
    printf("\nEnter the block number to read (0 to exit): ");  
    scanf("%d", &blockNumber);  
  
    if (blockNumber == 0)  
    {  
        break;  
    }  
    if (blockNumber < 0 || blockNumber >= 100 ||  
        indexBlock[blockNumber] == 0)  
    {  
        printf("Block not found.\n");  
    }  
    else  
    {  
        fseek(file, indexBlock[blockNumber], SEEK_SET);  
        fread(&block, sizeof(struct Block), 1, file);  
        printf("Block Number: %d\n", block.blockNumber);  
        printf("Data: %s\n", block.data);  
    }  
}  
  
fclose(file);  
return 0;  
}
```

Output:

```
Enter blocks (Enter '0' as block number to exit):  
Block Number: 2  
Data: 5  
Block Number: 5  
Data: 1  
Block Number: 8  
Data: 4  
Block Number: 7  
Data: 2  
Block Number: 2  
Data: 5  
Block Number: 1  
Data: 1  
Block Number: 1  
Data: 1  
Block Number: 1  
Data: 2  
Block Number: 0
```

```
Enter the block number to read (0 to exit): 0
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```